

Airline Flight Delay Prediction and Analysis

Phase 1 & Phase 2 Project Document

Course / Track	Advanced Data Analysis
Dataset	T_ONTIME_REPORTING(1).csv
Project Phases	1. Business Understanding 2. Data Understanding
Prepared for	Advanced Data Analysis Final Project

Document purpose

This document formalizes the first two phases of the final project roadmap: defining a clear business problem and establishing an evidence-based understanding of the dataset before data preparation and modelling.

1. Business Understanding

The first phase translates the pre-selected airline operations dataset into a clear, answerable business problem. The objective is to establish what the project is trying to solve, how success will be measured, and which analytical questions the project must answer.

1.1 Business Problem

Flight delays create operational and customer-service challenges for airlines and airports. A data-driven approach can help stakeholders identify patterns associated with late arrivals and support proactive operational planning.

Defined business problem

How can historical flight and operational information be used to understand delay patterns and predict whether an operating flight will arrive at least 15 minutes late?

1.2 Business Objective

The project aims to develop an analytical and machine-learning solution that:

- Identifies the flight, route, carrier, and operational characteristics associated with arrival delays.
- Provides clear visual evidence addressing key business questions about flight delays.
- Predicts whether an operating flight will arrive at least 15 minutes late.
- Compares alternative classification models and selects the model that best supports the defined business objective.
- Provides a foundation for deployment through an interactive Streamlit application in later project phases.

1.3 Target Variable

The proposed target variable is `ARR_DEL15`. It represents whether the flight arrived at least 15 minutes late and is therefore appropriate for framing the final modelling task as binary classification:

Value	Interpretation
0	Arrival delay of less than 15 minutes
1	Arrival delay of 15 minutes or more

1.4 Business Value

- Support earlier identification of flights with elevated delay risk.
- Improve operational awareness across airlines, airports, and route planning teams.
- Help prioritize attention to routes, carriers, and operating conditions associated with higher delay rates.
- Provide a practical predictive tool that can be demonstrated through a deployed application.

1.5 Success Criteria

The project should be evaluated using both business relevance and technical model performance.

Dimension	Measure	Project criterion
Business	Delay-risk identification	Produce interpretable findings that help explain which factors are associated with late arrivals.
Primary technical	F1-score	Target an F1-score of at least 0.70 on the held-out test set.
Supporting technical	Precision, Recall, Accuracy	Report and compare complementary classification metrics, with particular attention to recall for delayed flights.

Note: The F1-score threshold is a project target, not an observed result. Actual performance will only be established during modelling and evaluation.

1.6 Analytical Questions

1. Which airlines have the highest and lowest proportions of flights arriving 15 or more minutes late?
2. How is departure delay associated with the probability of arriving 15 or more minutes late?
3. Which origin and destination airports are associated with higher arrival-delay rates?
4. How do flight distance and scheduled flight characteristics relate to arrival delays?
5. Can arrival-delay status (15+ minutes late versus not) be predicted from information available about the flight?

1.7 Proposed Team Responsibilities

Workstream	Primary responsibility
Data Preparation	Data cleaning, missing values, outliers, encoding, scaling, feature engineering and train/test split.
EDA & Visualization	Charts, group comparisons, trends and written insights linked to the business questions.
Model Development	Model selection, training, feature preparation and model tuning.
Evaluation	Metric calculation, model comparison, confusion matrix and interpretation of results.
Deployment & Documentation	Streamlit application, GitHub repository, README and final presentation materials.

2. Data Understanding

The second phase profiles the dataset to understand its structure, quality, target distribution, and limitations before any modelling decisions are finalized.

2.1 Dataset Overview

Item	Finding
Records	544,003 rows
Variables	24 columns
Observed flight dates	01 January 2026 to 31 January 2026
Operating carriers	13 unique carriers
Origin airports	341 unique origin airports
Destination airports	342 unique destination airports
Origin states	52 unique states
Destination states	52 unique states

The dataset contains 24 variables covering flight dates, airline identifiers, airport and geographic information, scheduled times, departure and arrival delays, taxi-out time, cancellation information, actual elapsed time, and distance.

2.2 Variable Structure

Variable group	Representative columns
Flight identification	OP_UNIQUE_CARRIER, TAIL_NUM, OP_CARRIER_FL_NUM
Origin / destination	ORIGIN, ORIGIN_CITY_NAME, ORIGIN_STATE_NM, DEST, DEST_CITY_NAME, DEST_STATE_NM
Scheduled information	FL_DATE, CRS_DEP_TIME, CRS_ARR_TIME, ARR_TIME_BLK
Delay measures	DEP_DELAY, DEP_DEL15, ARR_DELAY, ARR_DEL15
Operational measures	TAXI_OUT, ACTUAL_ELAPSED_TIME, DISTANCE
Cancellation information	CANCELLED, CANCELLATION_CODE

2.3 Data Quality Assessment

The initial profiling produced the following quality findings:

- No duplicate rows were identified (0 duplicate rows).
- Missing values are concentrated in operational and delay-related fields, particularly CANCELLATION_CODE (518,368), ACTUAL_ELAPSED_TIME (26,781), ARR_DEL15 (26,781), ARR_DELAY (26,781), TAXI_OUT (25,515), DEP_DELAY (25,229), DEP_DEL15 (25,229) and TAIL_NUM (4,638).
- There are 25,635 cancelled records and 518,368 non-cancelled records.
- There are 517,222 records with a non-missing ARR_DEL15 outcome, allowing the classification target to be analysed after an appropriate treatment of cancelled/incomplete flights.

Variable	Missing values	Share of dataset
CANCELLATION_CODE	518,368	95.29%
ACTUAL_ELAPSED_TIME	26,781	4.92%
ARR_DEL15	26,781	4.92%
ARR_DELAY	26,781	4.92%
TAXI_OUT	25,515	4.69%
DEP_DELAY	25,229	4.64%
DEP_DEL15	25,229	4.64%
TAIL_NUM	4,638	0.85%

Interpretation: The large number of missing CANCELLATION_CODE values is consistent with the fact that most flights were not cancelled. Missing values in arrival and departure delay fields require additional investigation because they may be associated with cancelled or otherwise incomplete operations.

2.4 Target Distribution

ARR_DEL15	Meaning	Count	Share of non-missing target
0	Less than 15 minutes late	409,747	79.22%
1	15 minutes or more late	107,475	20.78%
—	Total non-missing ARR_DEL15	517,222	100.00%

Key modelling implication

The target is imbalanced but not extremely so: 20.8% of non-missing target observations are delayed and 79.2% are not delayed. Accuracy alone should therefore not be used to select the final model; F1-score, precision, recall and the confusion matrix should also be reviewed.

2.5 Cancellation Profile

Cancellation code	Count
B	22,463
A	2,246
C	879
D	47
Total coded cancellations	25,635

2.6 Numerical Range and Potential Outliers

Variable	Mean	Maximum
DEP_DELAY	13.10	3,339
ARR_DELAY	6.41	3,339
TAXI_OUT	19.40	213
ACTUAL_ELAPSED_TIME	146.30	763
DISTANCE	846.65	4,983

The delay variables contain extreme values, with both DEP_DELAY and ARR_DELAY reaching 3,339 minutes. These observations should not be removed automatically. During Phase 3, the team should determine whether extreme values represent valid operational events or data-quality issues and select an appropriate treatment.

2.7 Modelling Considerations and Data Limitations

- Temporal coverage is limited to the observed dates in the supplied file, from 1 January 2026 through 31 January 2026. Findings should therefore not be generalized to the entire year without additional evidence.
- The dataset does not contain direct weather variables, so weather-related causes of delay cannot be modelled directly from the available fields.
- Variables that become known only after or during a flight, such as ARR_DELAY and ACTUAL_ELAPSED_TIME, require careful treatment to avoid data leakage in an early prediction setting.
- ARR_DEL15 is missing for 26,781 records; the relationship between those missing outcomes and cancellations/incomplete operations should be investigated before the modelling dataset is finalized.
- Class imbalance means evaluation should emphasize metrics that distinguish performance on delayed flights, particularly recall and F1-score.

3. Phase 1–2 Consolidated Findings

Area	Consolidated finding
Business objective	Understand the drivers of airline arrival delays and predict whether an operating flight will arrive at least 15 minutes late.
Target	ARR_DEL15 as a binary classification target.
Dataset size	544,003 rows and 24 variables.
Time coverage	01 January 2026 to 31 January 2026.
Data quality	No duplicate rows; missingness is concentrated in cancellation and delay-related operational fields.
Target balance	79.2% non-delayed vs. 20.8% delayed among non-missing target observations.
Main risk	Potential leakage from variables only known after the flight, and potential bias from the limited time period.
Evaluation focus	F1-score as the primary target, supported by precision, recall, accuracy and confusion-matrix analysis.